Modeling Pyramidal Neuron Dynamics:

Comparing Normal & Epileptic Firing Rates with the Effects of Phenytoin & Ketamine

Abstract:

The goal of this experiment was to model and compare a normal pyramidal neuron to a pyramidal neuron affected by epilepsy and investigate the effects of the drugs phenytoin and ketamine on the epileptic model. My hypothesis is that phenytoin and ketamine will reduce the hyperexcitability of an epileptic neuron, bringing its firing behavior closer to that of a normal pyramidal neuron.

Understanding the specific mechanisms underlying hyperexcitability is essential for developing targeted therapeutic strategies to manage epilepsy. Computational modeling provides a controlled, reproducible platform to simulate these mechanisms. Such models are crucial for assessing the efficacy of existing drugs, understanding their mechanisms of action, and identifying areas for future research to design more effective treatments.

Epilepsy is a neurological disorder characterized by recurrent seizures and episodes of excessive electrical discharge. This condition affects a "normal neuronal network by turning it into a hyperexcitable network" (Bromfield, 2006). To study the effects of phenytoin and ketamine, a computational conductance - based model was first created to simulate a normal pyramidal neuron, serving as the control. This was followed by an epileptic version of the model. Simulations were subsequently run with the application of phenytoin and ketamine to observe their impact on neuronal behavior, phenytoin stabilizing sodium channel inactivation and ketamine blocking NMDA receptor activity (Gupta, 2023; Tan & Hashimoto, 2024). Spike count and membrane potential analysis provides quantitative measures of excitability, allowing for a clear assessment of how these drugs modulate epileptic firing dynamics.The results demonstrated that the normal neuron model fired 4 action potentials over 1000 ms, while the epileptic model showed an increased firing rate of 38 action potentials. Treatment with phenytoin and ketamine reduced the firing rates to 24 and 27 action potentials, respectively, indicating successful modulation of neuronal hyperexcitability. These findings support the hypothesis that these drugs will reduce the excitability in an epileptic neuron and bring its firing behavior closer to that of a normal pyramidal neuron.

Introduction:

To model the behavior of these neurons under both normal and epileptic conditions, a computational framework was developed that incorporates multiple ion channel conductances. This model integrates elements of the Hodgkin-Huxley (HH) model as well as some components of the Connors-Steven (CS) model to best reflect the unique electrophysiological properties of pyramidal neurons.

Sodium (Na^+) channels, regulated by activation (m) and inactivation (h) gating variables, are crucial for initiating and propagating action potentials. These channels open rapidly in response to depolarization, allowing a swift influx of Na^+ ions, and their subsequent inactivation terminates the depolarizing phase to restore the resting potential. Voltage-dependent rate constants (alpha m, beta m, alpha h, beta h) govern these kinetics, enabling the neuron’s characteristic fast-spiking behavior (Hodgkin & Huxley, 1952).

Potassium (K^+) channels, specifically the delayed rectifier type, are modeled with a single activation variable (n). These channels are critical for repolarization and maintaining the resting membrane potential. Their slower activation compared to sodium channels ensures that K^+ efflux follows Na^+-mediated depolarization, restoring the baseline state after an action potential(s). The voltage-dependent kinetics of potassium channels counteract sodium-driven depolarization, preventing prolonged excitation (Hodgkin & Huxley, 1952).

The model also incorporates calcium (Ca^{2+}) channels, which exhibit slower kinetics compared to sodium and potassium channels. (Qiao & Yao 2024). These channels activate in response to depolarization and allow Ca^{2+} influx, contributing to processes such as “cellular excitability, synaptic transmission and intracellular signaling pathways” (Ferron et al., 2021). The activation of calcium channels follows a sigmoidal voltage dependence with time constants reflecting their delayed response. (Qiao & Yao 2024).

Additionally, NMDA receptor-mediated currents “present at many excitatory glutamate synapses in the central nervous system” are included to capture the slow, prolonged excitatory effects of the current (Cull-Candy et al., 2001). These receptors, gated by voltage and ligand, exhibit a distinct dependence on the membrane potential, due to their blockade by extracellular magnesium ions (Cull-Candy et al., 2001). NMDA receptor kinetics are modeled with a steady-state gating variable and a fixed time constant, representing their slow response compared to other ion channels.

Leak channels, with constant conductance and reversal potential, allow non-specific ion flows that help stabilize the resting membrane potential.

Together, these ion channels govern the pyramidal neuron’s dynamic behavior. Sodium and potassium channels facilitate the rapid depolarization and repolarization phases of action potentials, while calcium and NMDA channels modulate slower, sustained depolarizations. This interplay of fast and slow conductances allows pyramidal neurons to exhibit a range of firing patterns, from single spikes to bursts, in response to synaptic inputs. The inclusion of these channels in the model provides a comprehensive framework for studying normal and pathological neuronal activity, particularly in the context of epilepsy, where hyperexcitability and excessive firing are hallmark features (Bromfield, 2006). This model offers a biologically grounded approach to exploring the modulation of epileptic activity by drugs such as phenytoin and ketamine, which target specific ion channel conductances to restore normal neuronal function.

Pyramidal Neuron (Control):

We first set parameters to initiate the core properties of the neuron model to in an attempt to accurately simulate a pyramidal neuron. In instances where values for a pyradmidal neuron were not available, values from the Hodgkin-Huxley Model were used. The membrane capacitance (C_m) of .75 was based off the model used by Mainen & Senowski acquired from ModelDB: demofig2.hoc (Mainen & Sejnowski, 1996). Parameters for leak conductance (G_leak) of 0.3 was based of maximum conductances found in the Hodkin-Huxley model (Hodgkin & Huxley, 1952). Calcium conductance (g_Ca_max) of 0.3 were based off maximum calcium conductance used by Mainen & Senowski acquired from ModelDB: demofig2.hoc.(Mainen & Sejnowski, 1996).Parameters for the reverse potentials for sodium (E_Na) of 50, potassium (E_K) of -90, and leak (E_leak) of -70 were based from the models used in Mainen & Sejnowski (Mainen & Sejnowski, 1996 pg 364).Parameters for calcium reversal (E_Ca) were set to 120 based off research by Zhou & Bers. (Zhou & Bers 2000). And lastly, the parameters for NMDA reversal (E_NMDA) were set to 0 mV based on research by Mayer

which found that NMDA receptors were permeable to Na, K, and Ca ions. Since the inward flow of Na & Ca is balanced by outward flow of K ions we can infer that the net reversal potential approximates to 0 mV (Mayer et al., 1984).

```
% Parameters for a Pyramidal Neuron

C_m = 0.75;         % Membrane capacitance (µF/cm^2)
g_Na_max = 100;     % Maximum sodium conductance (mS/cm^2)
g_K_max = 45;       % Maximum potassium conductance (mS/cm^2)
g_Ca_max = 0.3;     % Maximum calcium conductance (mS/cm^2)
g_NMDA = 0.15;      % NMDA receptor conductance (mS/cm^2)
G_leak = 0.3;       % Leak conductance (mS/cm^2)
E_Na = 50;          % Sodium reversal potential (mV)
E_K = -90;          % Potassium reversal potential (mV)
E_Ca = 120;         % Calcium reversal potential (mV)
E_NMDA = 0;         % NMDA reversal potential (mV)
E_leak = -70;       % Leak reversal potential (mV)
```

Next, we set up the time vector to place the bounds of our experiment, dictating the length and the detail of the simulation. We set the time step (dt) for this experiment to .001 (ms) to allow for a more precise view of the changes in membrane potential.

```
% Time Vector (ms)
dt = 0.001;
time = 0:dt:1000;
```

Next, we allocate a vector for the membrane potential (Vm) to store its value at each time step throughout the experiment. This is necessary to ensure the changes to the membrane potential are tracked and stored. Furthermore, allocating vectors allows for memory management and efficiency with the MATLAB program.

The initial membrane potential was set to -62mV typical of a pyramidal neuron as found by (Tsodyks, 1997).

```
% Allocate Vector for Membrane Potential (Vm)
Vm = zeros(1, length(time));
Vm(1) = -62;
```

Next, we intiate the resting gating variables. These values will get updated in each time step but to start, we have set the initiations at the neurons steady state of -62mV to ensure that the experiment begins from a realistic baseline. To acquire these values we calculate the alpha and beta rates at the initial membrane potential of -62mV and then input these values into the steady - state equation.

```
% Initialize Resting Gating Variables
m_Na = 0.0749;
h_Na = 0.4889;
n_Kd = 0.3645;
M_Ca = 0.3086;
NMDA_open = 0.2315;
```

Next, we set the applied current to induce spiking of the neuron so we can observe the responses of the model under variable circumstaces. For the model pyramidial neuron we set the initial applied current to 100. This will increase when we model an epileptic neuron.

```
% Applied Current (pA)
I_app = zeros(1, length(time));
I_app(time > 100 & time < 500) = 100;
```

Next, the for loop iterates through each time step of the simulation, dynamically updating the neuron's membrane potential and gating variables for each ion channel. At each step, the sodium, potassium, calcium, and NMDA ion channel kinetics are calculated based on their respective equations. The gating variables for sodium channels—activation (m) and inactivation (h)—are updated using Hodgkin-Huxley (HH) equations, which depend on the current membrane potential (Vm) (Beeman, 2014). Similarly, potassium channel activation (n) follows HH kinetics. Calcium and NMDA channel dynamics are modeled using slower kinetics derived from Connors, reflecting their role in sustaining depolarizations (Dayan & Abbott, 2001).

The membrane potential (Vm) is updated by summing all ionic currents (sodium, potassium, calcium, NMDA, and leak) and the applied current (Iapp). Each current contributes to the net ionic flow, with its magnitude determined by the channel's conductance, gating variables, and the driving force (the difference between Vm and the ion's reversal potential). The membrane potential is then incrementally adjusted to reflect these currents, ensuring a realistic simulation of the neuron's electrical activity.

```
% For Loop
for t = 1:length(time)-1

    % Sodium channel kinetics
    % Activation and inactivation variables based on HH kinetics
    alpha_m = (0.1 .* (Vm(t) + 40.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
40.0001)));
    beta_m = 4 .* exp(-0.0556 .* (Vm(t) + 65.0001));

    m_Na_inf = alpha_m / (alpha_m + beta_m);
    tau_m = 1 / (alpha_m + beta_m);

    % Inactivation variable h, rate constants follow HH kinetics
    alpha_h = 0.07 .* exp(-0.05 .* (Vm(t) + 65.0001));
    beta_h = 1 ./ (1 + exp(-0.1 .* (Vm(t) + 35.0001)));

    h_Na_inf = alpha_h / (alpha_h + beta_h);
    tau_h = 1 / (alpha_h + beta_h);

    % Potassium channel kinetics
    % Activation variable n follows HH kinetics
    alpha_n = (0.01 .* (Vm(t) + 55.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
55.0001)));
    beta_n = 0.125 .* exp(-0.0125 .* (Vm(t) + 65.0001));

    n_Kd_inf = alpha_n / (alpha_n + beta_n);
```

```
    tau_n = 1 / (alpha_n + beta_n);

    % Calcium channel kinetics
    % Activation variable M_Ca and time constant based on Connors
    M_Ca_inf = 1 / (1 + exp(-(Vm(t) + 57) / 6.2));
    tau_M_Ca = 0.612 + (1 / (exp(-(Vm(t) + 132) / 16.7) + exp((Vm(t) +
16.8) / 18.2)));

    % NMDA channel kinetics
    % Steady-state variable and slow time constant based on Connors
    NMDA_open_inf = 1 / (1 + exp(-(Vm(t) + 50) / 10));
    tau_NMDA = 50;  % Fixed slow response time constant

    % Update gating variables
    % Gating variables m, h, n (HH); M_Ca, NMDA_open (Connors)
    m_Na = m_Na_inf - (m_Na_inf - m_Na) * exp(-dt / tau_m);
    h_Na = h_Na_inf - (h_Na_inf - h_Na) * exp(-dt / tau_h);
    n_Kd = n_Kd_inf - (n_Kd_inf - n_Kd) * exp(-dt / tau_n);
    M_Ca = M_Ca_inf - (M_Ca_inf - M_Ca) * exp(-dt / tau_M_Ca);
    NMDA_open = NMDA_open_inf - (NMDA_open_inf - NMDA_open) * exp(-dt /
tau_NMDA);

    % Compute ionic currents
    % Sodium (I_Na) and Potassium (I_Kd) currents follow HH model
    % Calcium (I_Ca) and NMDA (I_NMDA) currents based on Connors
    % Leak current is generic, ensuring resting potential stability
    I_Na = g_Na_max * m_Na^3 * h_Na * (Vm(t) - E_Na);
    I_Kd = g_K_max * n_Kd^4 * (Vm(t) - E_K);
    I_Ca = g_Ca_max * M_Ca^2 * (Vm(t) - E_Ca);
    I_NMDA = g_NMDA * NMDA_open * (Vm(t) - E_NMDA);
    I_leak = G_leak * (Vm(t) - E_leak);

    % Update membrane potential
    % Membrane potential dynamics based on HH model
    dV = ((-I_Na - I_Kd - I_Ca - I_NMDA - I_leak + I_app(t)) / C_m) * dt;
    Vm(t+1) = Vm(t) + dV;
end
```

Next, we calculate the spike count to quantify the neuron's firing activity over the simulation. A spike is defined as a crossing of the threshold of -55mV, which is consistent with action potential initiation in pyramidal neurons (Mainen & Sejnowski, 1996). This ensures that only true spikes are counted, excluding subthreshold depolarizations or noise.

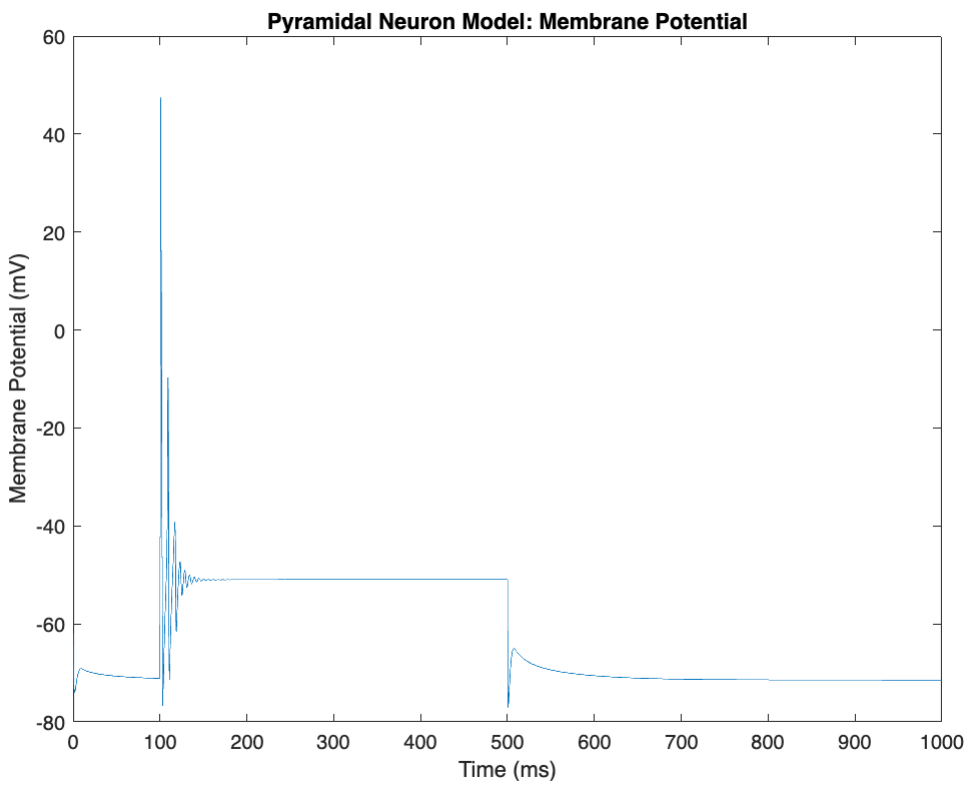
```
% Spike Count
spike_count = 0;
threshold = -55;

for t = 2:length(Vm)
    if Vm(t-1) < threshold && Vm(t) >= threshold
```

```
        spike_count = spike_count + 1;
    end
end

disp(['Spike Count: ', num2str(spike_count)]);
```

```
Spike Count: 4
```

Finally, we plot the correlation between Membrane Potential (mV) and Time (ms) to visually observe the firing rates of the model neuron.

```
% Plot Membrane Potential over Time
figure;
plot(time, Vm);
xlabel('Time (ms)');
ylabel('Membrane Potential (mV)');
title('Pyramidal Neuron Model: Membrane Potential');
```



In the model neuron, we observe that the membrane potential reaches a peak of approximately +50 mV during the first action potential, followed by subsequent peaks of diminishing amplitudes before stabilizing near the resting potential of -70mV. This behavior follows the findings of Colbert et al. which demonstrated that repetitive firing in pyramidal neurons leads to progressively diminished action potential amplitudes due to sodium channel inactivation (h) (Colbert et al., 1997).

Epileptic Neuron (Baseline Experimental Condition):

For the Epileptic Neuron Model we modified several parameters to simulate the hyperexcitability characteristic of epileptic pyramidal neurons. Modifications include increasing the sodium conductance (g_Na_max) to enhance rapid depolarization, as well as increasing calcium conductance (g_Ca_max) and NMDA conductance (g_NMDA) to promote prolonged depolarization. Additionally, we decreased potassium conductance (g_K_max) to slow repolarization and reduce inhibatory effects, and we lowered the leak conductance (G_leak) to further increase excitability. This parameter tuning allows the model to replicate the pathological dynamics observed in epilepsy, including prolonged bursts and reduced repolarization.

```
% Parameters for a Pyramidal Neuron with Epileptic Behavior
C_m = 0.75;
g_Na_max = 160;            % increased to enhance rapid depolarization
g_K_max = 18;              % decreased to slow repolarization and reduce
inhibatory effects
g_Ca_max = 0.4;            % increased to promote prolonged depolarization
g_NMDA = 0.25;             % increased for promote prolonged depolarization
G_leak = 0.1;              % reduced to further increase excitability
E_Na = 50;
E_K = -90;
E_Ca = 120;
E_NMDA = 0;
E_leak = -70;
```

The time vector, initial membrane potential, and resting gating variables remain unchanged.

```
% Time Vector (ms)
dt = 0.001;
time = 0:dt:1000;

% Allocate Vector for Membrane Potential (Vm)
Vm = zeros(1, length(time));
Vm(1) = -62;

% Initialize Resting Gating Variables
m_Na = 0.0749;
h_Na = 0.4889;
n_Kd = 0.3645;
M_Ca = 0.3086;
NMDA_open = 0.2315;
```

The applied current (Iapp) is increased to 200 pA and prolonged between 100 ms and 800 ms, mimicking the heightened and sustained excitatory input associated with epileptic behavior.

```
% Applied current (pA)
I_app = zeros(1, length(time));
I_app(time > 100 & time < 800) = 200;  % Stronger and prolonged applied
current
```

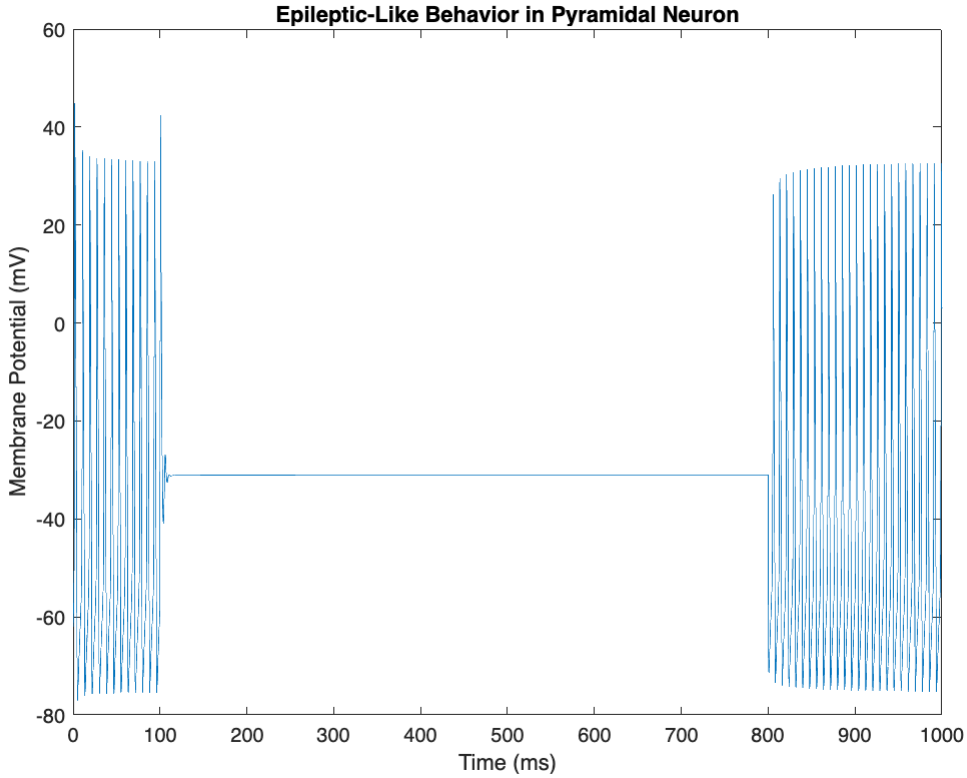
The loop is unchanged in the model epileptic neuron.

```
% For Loop
for t = 1:length(time)-1

    % Sodium channel kinetics
    % Activation and inactivation variables based on HH kinetics
    alpha_m = (0.1 .* (Vm(t) + 40.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
40.0001)));
    beta_m = 4 .* exp(-0.0556 .* (Vm(t) + 65.0001));

    m_Na_inf = alpha_m / (alpha_m + beta_m);
    tau_m = 1 / (alpha_m + beta_m);

    % Inactivation variable h, rate constants follow HH kinetics
    alpha_h = 0.07 .* exp(-0.05 .* (Vm(t) + 65.0001));
    beta_h = 1 ./ (1 + exp(-0.1 .* (Vm(t) + 35.0001)));

    h_Na_inf = alpha_h / (alpha_h + beta_h);
    tau_h = 1 / (alpha_h + beta_h);

    % Potassium channel kinetics
    % Activation variable n follows HH kinetics
    alpha_n = (0.01 .* (Vm(t) + 55.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
55.0001)));
    beta_n = 0.125 .* exp(-0.0125 .* (Vm(t) + 65.0001));

    n_Kd_inf = alpha_n / (alpha_n + beta_n);
    tau_n = 1 / (alpha_n + beta_n);

    % Calcium channel kinetics
    % Activation variable M_Ca and time constant based on Connors
    M_Ca_inf = 1 / (1 + exp(-(Vm(t) + 57) / 6.2));
    tau_M_Ca = 0.612 + (1 / (exp(-(Vm(t) + 132) / 16.7) + exp((Vm(t) +
16.8) / 18.2)));

    % NMDA channel kinetics
    % Steady-state variable and slow time constant based on Connors
    NMDA_open_inf = 1 / (1 + exp(-(Vm(t) + 50) / 10));
    tau_NMDA = 50;  % Fixed slow response time constant

    % Update gating variables
    % Gating variables m, h, n (HH); M_Ca, NMDA_open (Connors)
    m_Na = m_Na_inf - (m_Na_inf - m_Na) * exp(-dt / tau_m);
    h_Na = h_Na_inf - (h_Na_inf - h_Na) * exp(-dt / tau_h);
    n_Kd = n_Kd_inf - (n_Kd_inf - n_Kd) * exp(-dt / tau_n);
    M_Ca = M_Ca_inf - (M_Ca_inf - M_Ca) * exp(-dt / tau_M_Ca);
    NMDA_open = NMDA_open_inf - (NMDA_open_inf - NMDA_open) * exp(-dt /
tau_NMDA);

    % Compute ionic currents
    % Sodium (I_Na) and Potassium (I_Kd) currents follow HH model
```

```
    % Calcium (I_Ca) and NMDA (I_NMDA) currents based on Connors
    % Leak current is generic, ensuring resting potential stability
    I_Na = g_Na_max * m_Na^3 * h_Na * (Vm(t) - E_Na);
    I_Kd = g_K_max * n_Kd^4 * (Vm(t) - E_K);
    I_Ca = g_Ca_max * M_Ca^2 * (Vm(t) - E_Ca);
    I_NMDA = g_NMDA * NMDA_open * (Vm(t) - E_NMDA);
    I_leak = G_leak * (Vm(t) - E_leak);

    % Update membrane potential
    % Membrane potential dynamics based on HH model
    dV = ((-I_Na - I_Kd - I_Ca - I_NMDA - I_leak + I_app(t)) / C_m) * dt;
    Vm(t+1) = Vm(t) + dV;
end


% Spike Count
spike_count = 0;
threshold = -55;

for t = 2:length(Vm)
    if Vm(t-1) < threshold && Vm(t) >= threshold
        spike_count = spike_count + 1;
    end
end

disp(['Spike Count: ', num2str(spike_count)]);
```

```
Spike Count: 38
```

```
% Plot membrane potential over time
figure;
plot(time, Vm);
xlabel('Time (ms)');
ylabel('Membrane Potential (mV)');
title('Epileptic-Like Behavior in Pyramidal Neuron');
```



In the epileptic neuron model, the membrane potential plot shows significantly increased activity compared to the model neuron. The neuron fires approximately 38 action potentials, with bursts ranging from -75 mV to +35 mV. This heightened firing rate aligns with the hyperexcitability characteristic of epileptic neurons and demonstrates the expected increase in action potential generation compared to the spikes observed in the model neuron.

Phenytoin (Experimental Condition):

Phenytoin is an anticonvulsant drug commonly used to manage epilepsy by stabilizing neuronal membranes and reducing hyperexcitability. It achieves this primarily by targeting and blocking voltage-gated sodium channels, prolonging their inactivation state to limit repetitive firing and reduce the likelihood of sustained bursts (Gupta, 2003). In this model, the sodium conductance (g_Na_max) was reduced from 160 (epileptic baseline) to 120 to reflect this mechanism. Calcium conductance (g_Ca_max) was also reduced to 0.12 to account for phenytoin's secondary effects on reducing neurotransmitter release. Other parameters were kept consistent with the epileptic model, as phenytoin's primary action targets sodium and calcium channels.

```
% Parameters for an Epileptic Pyramidal Neuron with Phenytoin
C_m = 0.75;
g_Na_max = 120;           % reduced to limit hyperexcitability
g_K_max = 18;
g_Ca_max = 0.12;          % reduced to suppress prolonged depolarization
g_NMDA = 0.25;
G_leak = 0.1;
E_Na = 50;
```

```
E_K = -90;
E_Ca = 120;
E_NMDA = 0;
E_leak = -70;
```

The time vector, initial membrane potential, and resting gating variables remain unchanged.

```
% Time Vector (ms)
dt = 0.001;
time = 0:dt:1000;

% Initial conditions
Vm = zeros(1, length(time));
Vm(1) = -65;

% Gating variable initializations
m_Na = 0.0749;
h_Na = 0.4889;
n_Kd = 0.3645;
M_Ca = 0.3086;
NMDA_open = 0.2315;
```

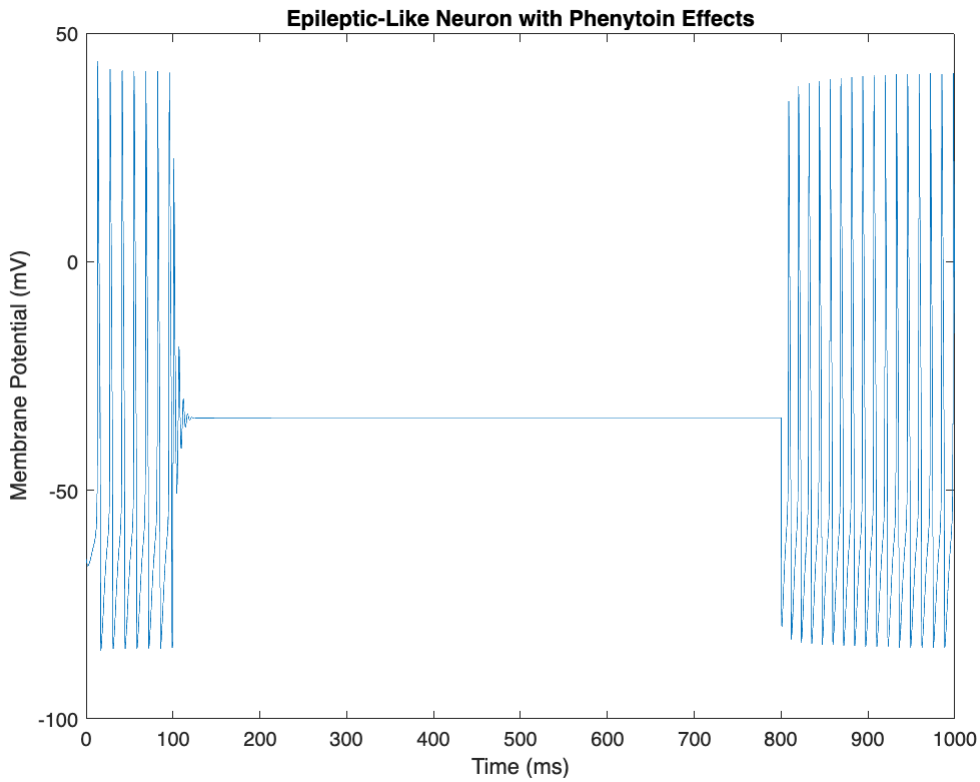
In the phenytoin-treated neuron model, the sodium inactivation time constant (tau_h) was increased by a factor of 1.5 to reflect phenytoin's ability to stabilize the inactivated state of sodium channels, thereby reducing repetitive firing. Additionally, the sodium current I_Na is affected due to the reduction of g_Na_max.

```
% Applied current (pA)
I_app = zeros(1, length(time));
I_app(time > 100 & time < 800) = 200;

% For Loop
for t = 1:length(time)-1

    % Sodium channel kinetics
    alpha_m = (0.1 .* (Vm(t) + 40.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
40.0001)));
    beta_m = 4 .* exp(-0.0556 .* (Vm(t) + 65.0001));

    m_Na_inf = alpha_m / (alpha_m + beta_m);
    tau_m = 1 / (alpha_m + beta_m);

    % Sodium inactivation kinetics with phenytoin adjustment
    alpha_h = 0.07 .* exp(-0.05 .* (Vm(t) + 65.0001));
    beta_h = 1 ./ (1 + exp(-0.1 .* (Vm(t) + 35.0001)));
    h_Na_inf = alpha_h / (alpha_h + beta_h);
    tau_h = (1 / (alpha_h + beta_h)) * 1.5;  % Increased inactivation time
constant to mimic phenytoin's effects

    % Potassium channel kinetics
```

```
    alpha_n = (0.01 .* (Vm(t) + 55.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
55.0001)));
    beta_n = 0.125 .* exp(-0.0125 .* (Vm(t) + 65.0001));
    n_Kd_inf = alpha_n / (alpha_n + beta_n);
    tau_n = 1 / (alpha_n + beta_n);

    % Calcium channel kinetics
    M_Ca_inf = 1 / (1 + exp(-(Vm(t) + 57) / 6.2));
    tau_M_Ca = 0.612 + (1 / (exp(-(Vm(t) + 132) / 16.7) + exp((Vm(t) +
16.8) / 18.2)));

    % NMDA channel kinetics
    NMDA_open_inf = 1 / (1 + exp(-(Vm(t) + 50) / 10));
    tau_NMDA = 50;

    % Update gating variables
    m_Na = m_Na_inf - (m_Na_inf - m_Na) * exp(-dt / tau_m);
    h_Na = h_Na_inf - (h_Na_inf - h_Na) * exp(-dt / tau_h);
    n_Kd = n_Kd_inf - (n_Kd_inf - n_Kd) * exp(-dt / tau_n);
    M_Ca = M_Ca_inf - (M_Ca_inf - M_Ca) * exp(-dt / tau_M_Ca);
    NMDA_open = NMDA_open_inf - (NMDA_open_inf - NMDA_open) * exp(-dt /
tau_NMDA);

    % Compute ionic currents
    I_Na = g_Na_max * m_Na^3 * h_Na * (Vm(t) - E_Na);  % Reduced Na+
conductance due reduced g_Na_max
    I_Kd = g_K_max * n_Kd^4 * (Vm(t) - E_K);
    I_Ca = g_Ca_max * M_Ca^2 * (Vm(t) - E_Ca);
    I_NMDA = g_NMDA * NMDA_open * (Vm(t) - E_NMDA);
    I_leak = G_leak * (Vm(t) - E_leak);

    % Update membrane potential
    dV = ((-I_Na - I_Kd - I_Ca - I_NMDA - I_leak + I_app(t)) / C_m) * dt;
    Vm(t+1) = Vm(t) + dV;
end


% Spike Count
spike_count = 0;
threshold = -55;

for t = 2:length(Vm)
    if Vm(t-1) < threshold && Vm(t) >= threshold
        spike_count = spike_count + 1;
    end
end

disp(['Spike Count: ', num2str(spike_count)]);
```

```
Spike Count: 24
```

```
% Plot membrane potential over time
figure;
plot(time, Vm);
xlabel('Time (ms)');
ylabel('Membrane Potential (mV)');
title('Epileptic-Like Neuron with Phenytoin Effects');
```



In the phenytoin-treated neuron model, the membrane potential plot shows reduced activity compared to the untreated epileptic neuron. The neuron fires a total of 24 action potentials, with spikes ranging from approximately -80 mV to +40 mV. The wider range of membrane potentials reflects phenytoin's ability to stabilize sodium channels, allowing for more complete action potential recovery and larger individual spikes while reducing the overall firing rate. This behavior aligns with phenytoin's role in mitigating hyperexcitability by suppressing repetitive firing.

Ketamine (Experimental Condition):

For our next comparison, we simulate the effects of ketamine on an epileptic neuron. Ketamine is an anesthetic and NMDA receptor antagonist (Zorumski et al., 2016). By blocking NMDA receptor activation, ketamine reduces calcium influx through these receptors, which helps restore the balance between excitatory and inhibitory (E/I) activities (Tan & Hashimoto, 2024). This mechanism is particularly relevant in epilepsy, where "abnormal expression or function" of NMDA receptors contributes to neuronal hyperexcitability and abnormal firing patterns (Sivakumar et al., 2002). In this model, the NMDA conductance was reduced from 0.25 to 0.1 to simulate ketamine's inhibitory effects on NMDA receptor activity.

```
% Parameters for Epileptic Pyramidal Neuron with Ketamine Effects
C_m = 0.75;
g_Na_max = 160;
g_K_max = 18;
g_Ca_max = 0.2;
g_NMDA = 0.1;              % Reduced NMDA conductance
G_leak = 0.1;
E_Na = 50;
E_K = -90;
E_Ca = 120;
E_NMDA = 0;
E_leak = -70;
```

The time vector, initial membrane potential, and resting gating variables remain unchanged

```
% Time step and simulation time (ms)
dt = 0.001;
time = 0:dt:1000;

% Initial conditions
Vm = zeros(1, length(time));
Vm(1) = -65;

% Gating variable initializations
m_Na = 0.0749;
h_Na = 0.4889;
n_Kd = 0.3645;
M_Ca = 0.3086;
NMDA_open = 0.2315;
```

In this section of the model, the effects of ketamine on neuronal activity are altered by the modified the NMDA receptor conductance (g_NMDA) parameter. This influences a reduction in the NMDA current (I_NMDA), which decreases the overall depolarizing current contributed by NMDA receptor activity. Consequently, this reduction propagates to the membrane potential update equation, resulting in a smaller change in membrane potential (dV) over time.

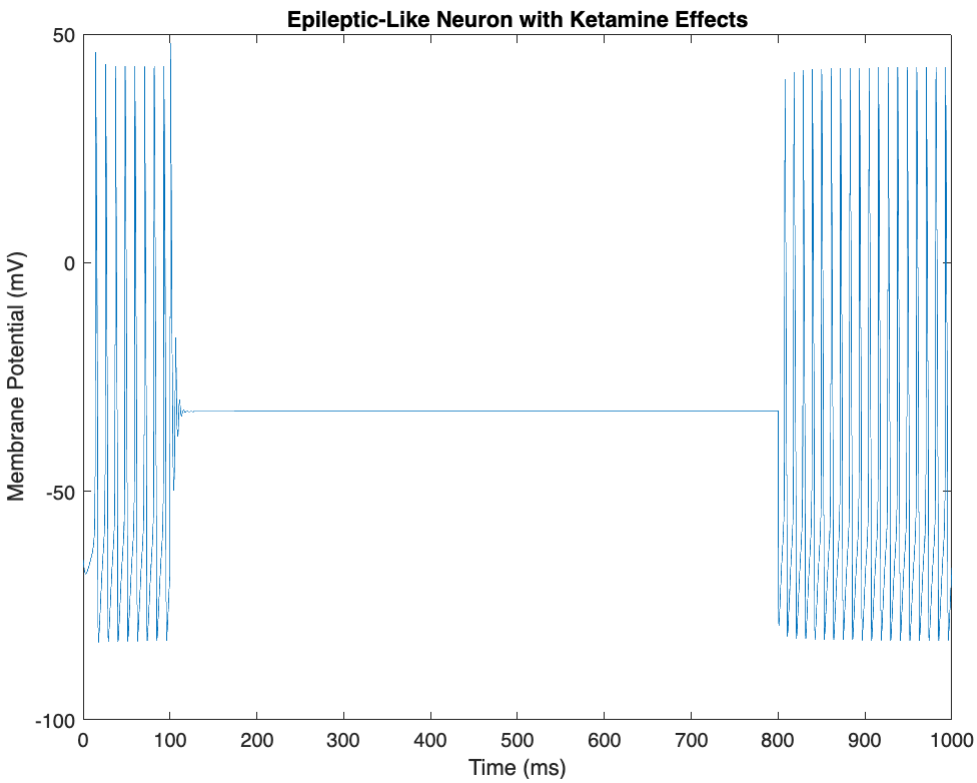
```
% Applied current (pA)
I_app = zeros(1, length(time));
I_app(time > 100 & time < 800) = 200;

% For Loop
for t = 1:length(time)-1

    % Sodium channel kinetics
    alpha_m = (0.1 .* (Vm(t) + 40.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
40.0001)));
    beta_m = 4 .* exp(-0.0556 .* (Vm(t) + 65.0001));
    m_Na_inf = alpha_m / (alpha_m + beta_m);
    tau_m = 1 / (alpha_m + beta_m);
```

```
    alpha_h = 0.07 .* exp(-0.05 .* (Vm(t) + 65.0001));
    beta_h = 1 ./ (1 + exp(-0.1 .* (Vm(t) + 35.0001)));
    h_Na_inf = alpha_h / (alpha_h + beta_h);
    tau_h = 1 / (alpha_h + beta_h);

    % Potassium channel kinetics
    alpha_n = (0.01 .* (Vm(t) + 55.0001)) ./ (1 - exp(-0.1 .* (Vm(t) +
55.0001)));
    beta_n = 0.125 .* exp(-0.0125 .* (Vm(t) + 65.0001));
    n_Kd_inf = alpha_n / (alpha_n + beta_n);
    tau_n = 1 / (alpha_n + beta_n);

    % Calcium channel kinetics
    M_Ca_inf = 1 / (1 + exp(-(Vm(t) + 57) / 6.2));
    tau_M_Ca = 0.612 + (1 / (exp(-(Vm(t) + 132) / 16.7) + exp((Vm(t) +
16.8) / 18.2)));

    % NMDA channel kinetics
    NMDA_open_inf = 1 / (1 + exp(-(Vm(t) + 50) / 10));
    tau_NMDA = 50;

    % Update gating variables
    m_Na = m_Na_inf - (m_Na_inf - m_Na) * exp(-dt / tau_m);
    h_Na = h_Na_inf - (h_Na_inf - h_Na) * exp(-dt / tau_h);
    n_Kd = n_Kd_inf - (n_Kd_inf - n_Kd) * exp(-dt / tau_n);
    M_Ca = M_Ca_inf - (M_Ca_inf - M_Ca) * exp(-dt / tau_M_Ca);
    NMDA_open = NMDA_open_inf - (NMDA_open_inf - NMDA_open) * exp(-dt /
tau_NMDA);

    % Compute ionic currents
    I_Na = g_Na_max * m_Na^3 * h_Na * (Vm(t) - E_Na);
    I_Kd = g_K_max * n_Kd^4 * (Vm(t) - E_K);
    I_Ca = g_Ca_max * M_Ca^2 * (Vm(t) - E_Ca);
    I_NMDA = g_NMDA * NMDA_open * (Vm(t) - E_NMDA);  % Reduced NMDA current
due to ketamine
    I_leak = G_leak * (Vm(t) - E_leak);

    % Update membrane potential
    dV = ((-I_Na - I_Kd - I_Ca - I_NMDA - I_leak + I_app(t)) / C_m) * dt;
    Vm(t+1) = Vm(t) + dV;
end


% Spike Count
spike_count = 0;
threshold = -55;

for t = 2:length(Vm)
    if Vm(t-1) < threshold && Vm(t) >= threshold
```

```
        spike_count = spike_count + 1;
    end
end

disp(['Spike Count: ', num2str(spike_count)]);
```

```
Spike Count: 27
```

```
% Plot membrane potential over time
figure;
plot(time, Vm);
xlabel('Time (ms)');
ylabel('Membrane Potential (mV)');
title('Epileptic-Like Neuron with Ketamine Effects');
```



In the ketamine-treated neuron model, the membrane potential plot shows reduced activity compared to the untreated epileptic neuron. The neuron fires a total of 27 action potentials, with bursts ranging from -70 mV to +45 mV. This reduction in firing rate reflects ketamine's ability to dampen NMDA receptor activity, mitigating hyperexcitability and abnormal firing patterns characteristic of epileptic neurons.

Conclusion:

This project modeled and analyzed the firing dynamics of a pyramidal neuron under four conditions: normal, epileptic, treated with phenytoin and treated with ketamine. The baseline neuron exhibited a well-defined first spike, with subsequent spikes diminishing in amplitude. While, the epileptic neuron demonstrated

hyperexcitability, characterized by rapid and sustained firing due to increased sodium, calcium, and NMDA conductances and reduced potassium and leak conductances. Phenytoin effectively reduced this hyperexcitability by stabilizing sodium channel inactivation and lowering sodium conductance, resulting in fewer spikes. Ketamine, an NMDA receptor antagonist, reduced NMDA receptor-mediated currents, bringing the firing closer to normal levels. In conclusion, the models of phenytoin and ketamine displayed a successful mitigation of epileptic activity by reducing firing rates, proving its efficacy as a form of treatment for epilepsy.

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